[2CEIT401 – Operating Systems]

Practical: 1

AIM- Hands on Practice of various Linux commands and analyse the output of each command.

Ganpat University
॥ विद्यया समाजोत्कर्षः ॥
U.V. Patel College of Engineering

Department of Computer Engineering

Aim: Hands on Practice of various Linux commands and analyse the output of each command.

1. TOUCH Command:

The touch command is used to create empty files. Multiple empty files can be created at once.

Syntax: touch <file_name> or touch <file1> <file2> ...

```
[24012011142@linuxserv ~]$ touch vatsal.txt
[24012011142@linuxserv ~]$ ls -l
total 4
drwxrwxr-x 2 24012011142 24012011142 4096 Jan  9 03:51 vatsal
-rw-rw-r-- 1 24012011142 24012011142    0 Jan 16 23:11 vatsal.txt
[24012011142@linuxserv ~]$
```

2. **CAT Command:**

The cat command is used to create, display, and copy file contents.

Syntax:

- To create a file: cat > <file_name>
- To display a file: cat <file_name>

```
[24012011142@linuxserv ~]$ cat > vatsal.txt
Hii My Name Is Vatsal

[5]+  Stopped                 cat >vatsal.txt
[24012011142@linuxserv ~]$ cat vatsal.txt
Hii My Name Is Vatsal
[24012011142@linuxserv ~]$
```

3. LS Command:

The ls command lists the contents of a directory.

Syntax: ls [flags]

```
[24012011142@linuxserv ~]$ ls -l
total 8
drwxrwxr-x 2 24012011142 24012011142 4096 Jan  9 03:51 vatsal
-rw-rw-r-- 1 24012011142 24012011142   22 Jan 16 23:17 vatsal.txt
[24012011142@linuxserv ~]$
```

4. MKDIR Command:

The mkdir command is used to create a new directory.

Syntax: mkdir <directory_name>

```
[24012011142@linuxserv ~]$ mkdir vatsal
[24012011142@linuxserv ~]$ ls -l
total 8
drwxrwxr-x 2 24012011142 24012011142 4096 Jan 16 23:21 vatsal
-rw-rw-r-- 1 24012011142 24012011142   22 Jan 16 23:17 vatsal.txt
[24012011142@linuxserv ~]$
```

5. RMDIR Command:

The rmdir command removes an empty directory.

Syntax: rmdir <directory_name>

```
[24012011142@linuxserv ~]$ ls -l
total 8
drwxrwxr-x 2 24012011142 24012011142 4096 Jan 16 23:21 vatsal
-rw-rw-r-- 1 24012011142 24012011142   22 Jan 16 23:17 vatsal.txt
[24012011142@linuxserv ~]$ rmdir vatsal
[24012011142@linuxserv ~]$ ls -l
total 4
-rw-rw-r-- 1 24012011142 24012011142 22 Jan 16 23:17 vatsal.txt
[24012011142@linuxserv ~]$
```

6. CD Command:

The cd command is used to change the current directory.

Syntax: cd <directory_name>

```
[24012011142@linuxserv ~]$ cd vatsal
[24012011142@linuxserv vatsal]$
```

7. CLEAR Command:

Clears the terminal screen.

Syntax: clear

```
[24012011142@linuxserv ~]$ clear
```

8. CP Command:

The cp command is used to copy files or directories.

Syntax: cp <source> <destination>

```
[24012011142@linuxserv ~]$ ls -l
total 8
drwxrwxr-x 2 24012011142 24012011142 4096 Jan 16 23:22 vatsal
-rw-rw-r-- 1 24012011142 24012011142   22 Jan 16 23:17 vatsal.txt
[24012011142@linuxserv ~]$ cp vatsal.txt vatsal
[24012011142@linuxserv ~]$ ls -l
total 8
drwxrwxr-x 2 24012011142 24012011142 4096 Jan 16 23:24 vatsal
-rw-rw-r-- 1 24012011142 24012011142   22 Jan 16 23:17 vatsal.txt
[24012011142@linuxserv ~]$ cd vatsal
[24012011142@linuxserv vatsal]$ ls -l
total 4
-rw-rw-r-- 1 24012011142 24012011142 22 Jan 16 23:24 vatsal.txt
[24012011142@linuxserv vatsal]$
```

9. CAL Command:

Displays the current month's calendar.

Syntax: cal [flags]

```
[24012011142@linuxserv ~]$ cal
    January 2026
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
```

10. HISTORY Command:

Displays previously executed commands.

Syntax: history

```
  139  ls -l
  140  cls
  141  cd ..
  142  cls
  143  cal
  144  cls
  145  history
  146  history
[24012011142@linuxserv ~]$
```

11. CHMOD Command:

Changes file permissions.

Syntax: chmod [permission] [file]

```
[24012011142@linuxserv ~]$ ls -l
total 8
drwxrwxr-x 2 24012011142 24012011142 4096 Jan 16 23:24 vatsal
-rw-rw-r-- 1 24012011142 24012011142   22 Jan 16 23:17 vatsal.txt
[24012011142@linuxserv ~]$ chmod 660 vatsal.txt
[24012011142@linuxserv ~]$ ls -l
total 8
drwxrwxr-x 2 24012011142 24012011142 4096 Jan 16 23:24 vatsal
-rw-rw---- 1 24012011142 24012011142   22 Jan 16 23:17 vatsal.txt
[24012011142@linuxserv ~]$
```

12. UMASK Command:

Sets default permissions for newly created files.

Syntax: umask [mask]

```
[24012011142@linuxserv ~]$ umask 007
[24012011142@linuxserv ~]$ touch test.txt
[24012011142@linuxserv ~]$ ls -l
total 8
-rw-rw---- 1 24012011142 24012011142    0 Jan 16 23:34 test.txt
drwxrwxr-x 2 24012011142 24012011142 4096 Jan 16 23:24 vatsal
-rw-rw-r-- 1 24012011142 24012011142   22 Jan 16 23:17 vatsal.txt
[24012011142@linuxserv ~]$
```

13. HEAD Command:

Displays the first 10 lines of a file.

Syntax: head <file name>

```
[24012011142@linuxserv ~]$ cat vatsal.txt
Hii I am Vatsal.
I am doing Computer Engineering.
line 3
line 4
line 5
line 6
[24012011142@linuxserv ~]$ head -n 5 vatsal.txt
Hii I am Vatsal.
I am doing Computer Engineering.
line 3
line 4
line 5
[24012011142@linuxserv ~]$
```

14. TAIL Command:

Displays the last 10 lines of a file.

Syntax: tail <file name>

```
[24012011142@linuxserv ~]$ cat vatsal.txt
Hii I am Vatsal.
I am doing Computer Engineering.
line 3
line 4
line 5
line 6
[24012011142@linuxserv ~]$ tail -n 5 vatsal.txt
I am doing Computer Engineering.
line 3
line 4
line 5
line 6
[24012011142@linuxserv ~]$
```

15. DATE Command:

Displays date, time, and time zone.

Syntax: date

```
[24012011142@linuxserv ~]$ date
Fri Jan 16 23:42:09 IST 2026
[24012011142@linuxserv ~]$
```

16. EXPR Command:

Evaluates expressions, including arithmetic and string operations.

Syntax: expr expression

```
[24012011142@linuxserv ~]$ expr 4 + 5
9
[24012011142@linuxserv ~]$ expr 4 - 5
-1
[24012011142@linuxserv ~]$ expr 4 \* 5
20
[24012011142@linuxserv ~]$ expr 20 / 5
4
[24012011142@linuxserv ~]$
```

17. WHO Command:

Shows information about currently logged-in users.

Syntax: who [options]

```
[24012011142@linuxserv ~]$ who
23012011085 pts/17       Jan 16 20:51 (152.59.34.106)
22012011065 pts/16       Jan 16 21:37 (142-156-103.vihaantelecom.com.142.156.103.in-addr.arpa)
23012011128 pts/5        Jan 16 21:50 (152.58.62.110)
23012011128 pts/18       Jan 16 21:59 (152.58.62.110)
24012011142 pts/1        Jan 16 23:10 (43.249.230.44)
23012011059 pts/2        Jan 16 23:45 (49.36.67.195)
[24012011142@linuxserv ~]$
```

18. UNAME Command:

Displays system information like OS name and kernel version.

Syntax: uname [options]

```
[24012011142@linuxserv ~]$ uname
Linux
[24012011142@linuxserv ~]$
```

19. FINGER Command:

Provides information about system users.

Syntax: finger [username]

```
[24012011142@linuxserv ~]$ finger
Login        Name       Tty      Idle  Login Time   Office     Office Phone
22012011065             pts/16   2:04  Jan 16 21:37 (142-156-103.vihaantelecom.com.142.156.103.in-addr.arpa)
23012011059             pts/2          Jan 16 23:45 (49.36.67.195)
23012011128             pts/5    1:56  Jan 16 21:50 (152.58.62.110)
23012011128             pts/18   1:46  Jan 16 21:59 (152.58.62.110)
24012011142             pts/1          Jan 16 23:10 (43.249.230.44)
[24012011142@linuxserv ~]$
```

20. CMP Command:

Compares two files byte by byte.

Syntax: cmp [file1] [file2]

```
[24012011142@linuxserv ~]$ cmp vatsal.txt test.txt
vatsal.txt test.txt differ: char 16, line 1
[24012011142@linuxserv ~]$
```

21. COMM Command:

Compares two files and displays non-matching/matching items.

Syntax: comm <file1> <file2>

```
[24012011142@linuxserv ~]$ comm vatsal.txt test.txt
        Hii I am Vatsal Patel.
Hii I am Vatsal.
                I am doing Computer Engineering.
                line 3
[24012011142@linuxserv ~]$
```

22. SORT Command:

Sorts files in alphabetical order.

Syntax: sort <file name>

```
[24012011142@linuxserv ~]$ cat vatsal.txt
Hii I am Vatsal.
I am doing Computer Engineering.
line 3
A
[24012011142@linuxserv ~]$ sort vatsal.txt
A
Hii I am Vatsal.
I am doing Computer Engineering.
line 3
[24012011142@linuxserv ~]$
```

23. SPELL Command:

Checks for spelling errors in a file.

Syntax: spell [file]

```
[24012011142@linuxserv ~]$ spell vatsal.txt
Hii
Vatsal
[24012011142@linuxserv ~]$
```

24. WC Command:

Counts lines, words, and characters in a file.

Syntax: wc <file name>

```
[24012011142@linuxserv ~]$ wc vatsal.txt
 4 12 59 vatsal.txt
[24012011142@linuxserv ~]$
```

25. TYPE Command:

Identifies the type of a command.

Syntax: type [command]

```
[24012011142@linuxserv ~]$ type cd
cd is a shell builtin
[24012011142@linuxserv ~]$ type rm
rm is hashed (/bin/rm)
[24012011142@linuxserv ~]$ type mkdir
mkdir is hashed (/bin/mkdir)
[24012011142@linuxserv ~]$ type type
type is a shell builtin
[24012011142@linuxserv ~]$
```

26. TTY Command:

Displays the terminal name.

Syntax: tty

```
[24012011142@linuxserv ~]$ tty
/dev/pts/1
[24012011142@linuxserv ~]$
```

27. ECHO Command:

Prints text or variables to the terminal.

Syntax: echo [string]

```
[24012011142@linuxserv ~]$ echo -e "Hii My Name is Vatsal. \nI am Doing Computer Engineering."
Hii My Name is Vatsal.
I am Doing Computer Engineering.
[24012011142@linuxserv ~]$
```

28. MAN Command:

Opens the manual page for a command.

Syntax: man [command]

```
man(1)                                                                  man(1)

NAME
       man - format and display the on-line manual pages

SYNOPSIS
       man  [-acdfFhkKtwW] [--path] [-m system] [-p string] [-C config_file] [-M pathlist] [-P pager] [-B browser] [-H htmlpager] [-S
       section_list] [section] name ...

DESCRIPTION
       man formats and displays the on-line manual pages.  If you specify section, man only looks in  that  section  of  the  manual.
       name  is  normally the name of the manual page, which is typically the name of a command, function, or file.  However, if name
       contains a slash (/) then man interprets  it  as  a  file  specification,  so  that  you  can  do  man  ./foo.5  or  even  man
       /cd/foo/bar.1.gz.

       See below for a description of where man looks for the manual page files.

OPTIONS
       -C  config_file
              Specify the configuration file to use; the default is /etc/man.config.  (See man.config(5).)

       -M  path
              Specify  the  list of directories to search for man pages.  Separate the directories with colons.  An empty list is the
              same as not specifying -M at all.  See SEARCH PATH FOR MANUAL PAGES.

       -P  pager
              Specify which pager to use.  This option overrides the MANPAGER environment variable, which in turn overrides the PAGER
              variable.  By default, man uses /usr/bin/less -is.

       -B     Specify  which  browser  to use on HTML files.  This option overrides the BROWSER environment variable. By default, man
              uses /usr/bin/less-is,

       -H     Specify a command that renders HTML files as text.  This  option  overrides  the  HTMLPAGER  environment  variable.  By
              default, man uses /bin/cat,
```

```
       -S  section_list
              List is a colon separated list of manual sections to search.  This option overrides the MANSECT environment variable.

       -a     By default, man will exit after displaying the first manual page it finds.  Using this option forces man to display all
              the manual pages that match name, not just the first.

       -c     Reformat the source man page, even when an up-to-date cat page exists.  This can be meaningful if the cat page was for-
              matted for a screen with a different number of columns, or if the preformatted page is corrupted.

       -d     Don't actually display the man pages, but do print gobs of debugging information.

       -D     Both display and print debugging info.

       -f     Equivalent to whatis.

       -F or --preformat
              Format only - do not display.

       -h     Print a help message and exit.

       -k     Equivalent to apropos.

       -K     Search for the specified string in *all* man pages. Warning: this is probably very slow! It helps to specify a section.
              (Just to give a rough idea, on my machine this takes about a minute per 500 man pages.)

       -m  system
              Specify an alternate set of man pages to search based on the system name given.

       -p  string
              Specify the sequence of preprocessors to run before nroff or troff.  Not all installations will have a full set of pre-
              processors.  Some of the preprocessors and the letters used to designate them are: eqn (e), grap (g), pic (p), tbl (t),
              vgrind (v), refer (r).  This option overrides the MANROFFSEQ environment variable.
```

29. MORE Command:

Displays file content one page at a time.

Syntax: more <file name>

```
[24012011142@linuxserv ~]$ man cd | more
```

```
[24012011142@linuxserv ~]$ more test.txt
```

```
BASH_BUILTINS(1)                                          BASH_BUILTINS(1)



NAME
       bash,  :,  ., [, alias, bg, bind, break, builtin, cd, command, compgen,
       complete, continue, declare, dirs, disown, echo,  enable,  eval,  exec,
       exit,  export,  fc,  fg, getopts, hash, help, history, jobs, kill, let,
       local, logout, popd, printf, pushd, pwd, read, readonly,  return,  set,
       shift,  shopt,  source,  suspend,  test,  times,  trap,  type, typeset,
       ulimit, umask, unalias, unset,  wait  -  bash  built-in  commands,  see
       bash(1)

BASH BUILTIN COMMANDS
       Unless otherwise noted, each builtin command documented in this section
       as accepting options preceded by - accepts -- to signify the end of the
       options.   For  example,  the  :, true, false, and test builtins do not
       accept options.
       : [arguments]
              No effect; the command does nothing beyond  expanding  arguments
              and  performing any specified redirections.  A zero exit code is
              returned.

        .  filename [arguments]
       source filename [arguments]
              Read and execute commands from filename  in  the  current  shell
              environment  and return the exit status of the last command exe-
              cuted from filename.  If filename does not contain a slash, file
              names  in  PATH  are used to find the directory containing file-
              name.  The file searched for in PATH  need  not  be  executable.
              When  bash  is  not  in  posix  mode,  the  current directory is
              searched if no file is found in PATH.  If the sourcepath  option
              to  the  shopt  builtin  command  is turned off, the PATH is not
              searched.  If any arguments are supplied, they become the  posi-
--More--(2%)
```

30. PASSWD Command:

Creates and changes the user password.

Syntax: passwd <username>

```
[24012011142@linuxserv ~]$ passwd
Changing password for user 24012011142.
Changing password for 24012011142
(current) UNIX password:
New UNIX password:
Retype new UNIX password:
passwd: all authentication tokens updated successfully.
[24012011142@linuxserv ~]$ |
```

31. PWD Command:

Displays the current working directory.

Syntax: pwd

```
[24012011142@linuxserv ~]$ pwd
/home/24012011142
[24012011142@linuxserv ~]$ |
```

32. GREP Command:

Searches for content within a file.

Syntax: command | grep <searchWord>

```
[24012011142@linuxserv ~]$ cat vatsal.txt | grep "Vatsal"
Hii I am Vatsal.
[24012011142@linuxserv ~]$ |
```

33. PS Command:

The ps command displays currently running processes.

Syntax: ps [options]

```
[24012011142@linuxserv ~]$ ps
  PID TTY          TIME CMD
13339 pts/1    00:00:00 bash
13508 pts/1    00:00:00 cat
13510 pts/1    00:00:00 cat
13594 pts/1    00:00:00 cat
13598 pts/1    00:00:00 cat
13604 pts/1    00:00:00 cat
14102 pts/1    00:00:00 cat
14105 pts/1    00:00:00 cat
14115 pts/1    00:00:00 cat
14120 pts/1    00:00:00 cat
14121 pts/1    00:00:00 cat
14122 pts/1    00:00:00 cat
14125 pts/1    00:00:00 cat
14567 pts/1    00:00:00 sort
14580 pts/1    00:00:00 cat
15421 pts/1    00:00:00 grep
15436 pts/1    00:00:00 ps
[24012011142@linuxserv ~]$
```

34. RM Command:

The rm command is used to remove a file.

Syntax: rm <file name>

```
[24012011142@linuxserv vatsal]$ ls -l
total 4
-rw-rw-r-- 1 24012011142 24012011142 16 Jan  1 18:50 v1
[24012011142@linuxserv vatsal]$ rm v1
[24012011142@linuxserv vatsal]$ ls -l
total 0
[24012011142@linuxserv vatsal]$
```

35. SET Command:

The set command is used to set shell variables and options.

Syntax: set [option]

```
MACHTYPE=i686-redhat-linux-gnu
MAIL=/var/spool/mail/24012011142
MAILCHECK=60
OLDPWD=/home/24012011142/vatsal
OPTERR=1
OPTIND=1
OSTYPE=linux-gnu
PATH=/usr/lib/qt-3.3/bin:/usr/kerberos/bin:/usr/java/jdk1.6.0_04/bin:/usr/local/bin:/bin:/usr/bin:/home/24012011142/bin
PIPESTATUS=([0]="0")
PPID=13338
PRELINKING=yes
PRELINK_FULL_TIME_INTERVAL=14
PRELINK_NONRPM_CHECK_INTERVAL=7
PRELINK_OPTS=-mR
PS1='[\u@\h \W]\$ '
PS2='> '
PS4='+ '
PWD=/home/24012011142
QTDIR=/usr/lib/qt-3.3
QTINC=/usr/lib/qt-3.3/include
QTLIB=/usr/lib/qt-3.3/lib
REMOTEHOST=43.249.230.44
SHELL=/bin/bash
SHELLOPTS=braceexpand:emacs:hashall:histexpand:history:interactive-comments:monitor
SHLVL=1
SSH_ASKPASS=/usr/libexec/openssh/gnome-ssh-askpass
TERM=ansi
UID=8906
USER=24012011142
_=clear
qt_prefix=/usr/lib/qt-3.3
[24012011142@linuxserv ~]$
```

36. CUT Command:

The cut command extracts specific columns from a file. The -d option specifies a delimiter - it can be a space (' '), a slash (/), a hyphen (-), or anything else, and the -f option selects a column.

Syntax: cut -d(delimiter) -f(columnNumber) <fileName>

```
[24012011142@linuxserv ~]$ who | cut -d ' ' -f 1
24012011142
23012011059
23012011128
23012011059
[24012011142@linuxserv ~]$
```

37. **READ Command:**

The read command reads user input and stores it in a variable.

Syntax: read [variable]

```
[24012011142@linuxserv ~]$ read a
Vatsal
[24012011142@linuxserv ~]$ echo "$a"
Vatsal
[24012011142@linuxserv ~]$
```

38. **JOBS Command:**

Lists active background jobs in the current shell.

Syntax: jobs [options]

```
[24012011142@linuxserv ~]$ jobs
[1]   Stopped                 cat >vatsal.txt
[2]   Stopped                 cat >vatsal.txt
[3]   Stopped                 cat >vatsal.txt
[4]   Stopped                 cat >vatsal.txt
[5]   Stopped                 cat >vatsal.txt
[6]   Stopped                 cat >>vatsal.txt
[7]   Stopped                 cat >>vatsal.txt
[8]   Stopped                 cat >>vatsal.txt
[9]   Stopped                 cat >>vatsal.txt
[10]   Stopped                 cat >>vatsal.txt
[11]   Stopped                 cat >>vatsal.txt
[12]   Stopped                 cat >>vatsal.txt
[13]   Stopped                 sort >>vatsal.txt
[14]-  Stopped                 cat >>vatsal.txt
[15]+  Stopped                 grep vatsal
[24012011142@linuxserv ~]$
```

39. AWK Command:

awk is a powerful text-processing and pattern-matching tool.

Syntax: awk 'pattern {action}' file

```
[24012011142@linuxserv ~]$ cat data.txt
142 Vatsal 20 CE
150 Aryan 21 IT
103 Aman 22 CE
[24012011142@linuxserv ~]$ awk '{ print $2, $4 }' data.txt
Vatsal CE
Aryan IT
Aman CE
[24012011142@linuxserv ~]$
```

40. **LN Command:**

The ln command creates hard or symbolic (soft) links to files.

Syntax: ln [options] target link_name

```
[24012011142@linuxserv ~]$ cat vatsal1/test.txt
101 Vatsal 20 CE
102 Rahul 21 IT
103 Aman 22 CS

[24012011142@linuxserv ~]$ ls -l
total 12
-rw-rw---- 1 24012011142 24012011142   48 Jan 17 00:40 data.txt
-rw-rw-r-- 1 24012011142 24012011142   59 Jan 16 23:55 vatsal.txt
drwxrwxr-x 2 24012011142 24012011142 4096 Jan 17 00:47 vatsal1
[24012011142@linuxserv ~]$  ln vatsal1/test.txt tes
[24012011142@linuxserv ~]$ cat tes
101 Vatsal 20 CE
102 Rahul 21 IT
103 Aman 22 CS

[24012011142@linuxserv ~]$
```

41. **ENV Command:**

The env displays all environment variables or runs a command with a modified environment.

Syntax: env [options] ENVName=ENVValue

```
[24012011142@linuxserv ~]$ env NAME=Vatsal bash
[24012011142@linuxserv ~]$ echo $NAME
Vatsal
[24012011142@linuxserv ~]$ |
```

```
[24012011142@linuxserv ~]$ env
REMOTEHOST=43.249.230.44
HOSTNAME=linuxserv
SHELL=/bin/bash
TERM=ansi
HISTSIZE=1000
KDE_NO_IPV6=1
QTDIR=/usr/lib/qt-3.3
QTINC=/usr/lib/qt-3.3/include
USER=24012011142
LS_COLORS=no=00:fi=00:di=01;34:ln=01;36:pi=40;33:so=01;35:bd=40;33;01:cd=40;33;01:or=01;05;37;41:mi=01;05;37;41:ex=01;32:*.cmd=01;32:*.exe=01;32:*.c
om=01;32:*.btm=01;32:*.bat=01;32:*.sh=01;32:*.csh=01;32:*.tar=01;31:*.tgz=01;31:*.arj=01;31:*.taz=01;31:*.lzh=01;31:*.zip=01;31:*.z=01;31:*.Z=01;31:
*.gz=01;31:*.bz2=01;31:*.bz=01;31:*.tz=01;31:*.rpm=01;31:*.cpio=01;31:*.jpg=01;35:*.gif=01;35:*.bmp=01;35:*.xbm=01;35:*.xpm=01;35:*.png=01;35:*.tif=
01;35:
KDEDIR=/usr
MAIL=/var/spool/mail/24012011142
PATH=/usr/lib/qt-3.3/bin:/usr/kerberos/bin:/usr/java/jdk1.6.0_04/bin:/usr/local/bin:/bin:/usr/bin:/home/24012011142/bin
INPUTRC=/etc/inputrc
PWD=/home/24012011142
KDE_IS_PRELINKED=1
SSH_ASKPASS=/usr/libexec/openssh/gnome-ssh-askpass
SHLVL=1
HOME=/home/24012011142
LOGNAME=24012011142
QTLIB=/usr/lib/qt-3.3/lib
CVS_RSH=ssh
LESSOPEN=|/usr/bin/lesspipe.sh %s
G_BROKEN_FILENAMES=1
_=/bin/env
OLDPWD=/home/24012011142/vatsal1
[24012011142@linuxserv ~]$ |
```

42. **KILL Command:**

The kill command terminates a process using its Process ID (PID).

Syntax: kill [signal] PID

```
[24012011142@linuxserv ~]$ ./wait.bash &
[1] 16244
[24012011142@linuxserv ~]$ Waiting process started. PID = 16244

[24012011142@linuxserv ~]$ ps
  PID TTY          TIME CMD
15833 pts/1    00:00:00 bash
16067 pts/1    00:00:00 bash
16244 pts/1    00:00:00 wait.bash
16246 pts/1    00:00:00 sleep
16247 pts/1    00:00:00 ps
[24012011142@linuxserv ~]$ kill 16244
[24012011142@linuxserv ~]$ ps
  PID TTY          TIME CMD
15833 pts/1    00:00:00 bash
16067 pts/1    00:00:00 bash
16261 pts/1    00:00:00 ps
[1]+  Terminated              ./wait.bash
[24012011142@linuxserv ~]$
```

43. **ALIAS Command:**

The alias command defines shortcuts for longer commands.

Syntax: alias name='command'

```
[24012011142@linuxserv ~]$ alias ls='ls -l'
[24012011142@linuxserv ~]$ ls
total 16
-rw-rw---- 1 24012011142 24012011142   48 Jan 17 00:40 data.txt
-rw-rw-r-- 1 24012011142 24012011142   59 Jan 16 23:55 vatsal.txt
drwxrwxr-x 2 24012011142 24012011142 4096 Jan 17 00:47 vatsal1
-rwxrw-r-- 1 24012011142 24012011142   87 Jan 17 01:01 wait.bash
[24012011142@linuxserv ~]$
```

44. **DIFF Command:**

The diff command compares two files line by line and shows differences.

Syntax: diff [file1] [file2]

```
[24012011142@linuxserv ~]$ cat vatsal.txt
Hii I am Vatsal.
I am doing Computer Engineering.
line 3
[24012011142@linuxserv ~]$ cat vatsal1.txt
Hii I am Vatsal.
I am doing Computer Engineering.
line 3
A
[24012011142@linuxserv ~]$ diff vatsal.txt vatsal1.txt
3a4
> A
[24012011142@linuxserv ~]$
```

45. **LOCATE Command:**

The locate command searches for a file by its name. Unlike find, it works in the background using a database, making it faster.

Syntax: locate <file name>

```
[24012011142@linuxserv ~]$ locate bashrc
/etc/bashrc
/etc/skel/.bashrc
/home/22012571019/.bashrc
/home/24012011142/.bashrc
/usr/share/doc/bash-3.1/startup-files/bashrc
[24012011142@linuxserv ~]$
```

```
[24012011142@linuxserv ~]$ locate bash_profile
/etc/skel/.bash_profile
/home/22012571019/.bash_profile
/home/24012011142/.bash_profile
[24012011142@linuxserv ~]$
```

46. **FIND Command:**

The find command searches for files in directories. It supports searching by name, type, date, and more.

Symbols Used in find Command:

- (.) → Current directory
- (/) → Root directory

Syntax: find . -name "*.pdf"

```
[24012011142@linuxserv ~]$ find . -name "*.txt"
./vatsal1.txt
./data.txt
./vatsal.txt
[24012011142@linuxserv ~]$
```

47. INFO Command:

The info command provides detailed documentation about a command (alternative to man).

Syntax: info [command]

```
File: *manpages*,  Node: cd,  Up: (dir)

BASH_BUILTINS(1)                                                BASH_BUILTINS(1)

NAME
       bash,  :,  .,  [, alias, bg, bind, break, builtin, cd, command, compgen,
       complete, continue, declare, dirs, disown, echo,  enable,  eval,  exec,
       exit,  export,  fc,  fg, getopts, hash, help, history, jobs, kill, let,
       local, logout, popd, printf, pushd, pwd, read, readonly,  return,  set,
       shift,  shopt,  source,  suspend,  test,  times,  trap,  type, typeset,
       ulimit, umask, unalias, unset,  wait  -  bash  built-in  commands,  see
       bash(1)

BASH BUILTIN COMMANDS
       Unless otherwise noted, each builtin command documented in this section
       as accepting options preceded by - accepts -- to signify the end of the
       options.   For  example,  the  :, true, false, and test builtins do not
       accept options.
       : [arguments]
              No effect; the command does nothing beyond  expanding  arguments
              and  performing any specified redirections.  A zero exit code is
              returned.

        .  filename [arguments]
       source filename [arguments]
              Read and execute commands from filename  in  the  current  shell
              environment  and return the exit status of the last command exe-
              cuted from filename.  If filename does not contain a slash, file
              names  in  PATH  are used to find the directory containing file-
              name.  The file searched for in PATH  need  not  be  executable.
              When  bash  is  not  in  posix  mode,  the  current directory is
              searched if no file is found in PATH.  If the sourcepath  option
              to  the  shopt  builtin  command  is turned off, the PATH is not
-----Info: (*manpages*)cd, 1407 lines --Top-----------------------------------
Welcome to Info version 4.8. Type ? for help, m for menu item.
```